WHITE PAPER · PORTFOLIO AND AI GOVERNANCE

The winning AI proposal tends to fall short, even when no one oversold

Ranking by estimates favors proposals whose estimates came out high, more so in a crowded round and where the evidence is thinnest. So move each claim toward what such proposals deliver before you rank.



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THE CENTRAL ARGUMENT

When competing AI proposals are ranked by estimated value and funded from the top, the winners tend to deliver less than they claimed. That holds even when every sponsor is honest, because ranking favors the estimates that happened to come out high. The gap grows with the number of close rivals. And a thinly evidenced winner's lead is more likely to be luck. A cut of the same share to every claim cannot change the winner. Moving each claim toward a typical figure, by more where the evidence is thin, can change it. That move also changes how a shortfall should be read.

Several AI proposals want the same people and budget. Each team fills in the same template and states the value it expects. Your review ranks them by a weighted score, or by value for the effort. It funds from the top until the capacity runs out. You already allow for two problems. Sponsors oversell, so the review trims claimed benefits or sets a higher bar. And some ideas rest on thin evidence, so the review sets a higher bar for newer kinds of work or scores confidence. The RICE method, for example, multiplies each idea's score by 100, 80 or 50 percent for high, medium or low confidence [1]. Underneath sits a reasonable belief. If every estimate were honest, funding the top of the list would deliver what the list says.

It would not. Even when every sponsor is honest, the proposals that reach the top are more likely to be ones whose estimates happened to come out high. So the ones you fund should be expected, on average, to deliver less than they claim. The gap grows with the number of close rivals in the round. And when a proposal with thin evidence wins, more of its lead is likely to be luck in its estimate. For the same claim, it should be expected to fall further short than a well-measured winner.

Two things follow that the usual discounts miss. First, cutting every claim by the same share can shrink the gap you expect. But it cannot change which proposal wins. Second, when a funded proposal falls short of its claim, that alone does not show its sponsor oversold. The ranking produces a shortfall by itself.

So before the review reads the cases, the chair writes down a typical figure for this kind of proposal and the range they would expect. Each claim is then moved toward that typical figure. The less its evidence narrows the range, the further it moves. Rank and fund on the moved figures. Record them at approval, and judge delivery against them.

ARGUMENT AT A GLANCE

What a ranking of AI proposals should compare

What the ranking does

Favors estimates that came out high, so winners tend to fall short even when honest.

What the usual cut misses

A same-share cut keeps the same winner; tiers pull toward zero.

What to rank on

Each claim moved toward a typical figure, keeping more where the evidence is stronger.

Applies to proposals competing for the same capacity and compared on estimated value or a score built from it.

Why honest estimates still disappoint

In 2006 James Smith and Robert Winkler of Duke University explained why a careful decision maker tends to feel let down after following good analysis [2]. They looked at the most ordinary way of choosing: estimate the value of each option, then pick the highest. They assumed every estimate was honest, meaning right on average for that option. Each estimate is the true value plus some error, as likely to be high as low.

The trouble is in the picking. When you rank on the estimate, you are ranking on true value plus error. The option that comes out on top is more likely to be one whose error happened to be high. So its estimate is, on average, too high. Smith and Winkler proved that this holds whenever there is any chance of picking the wrong option, whatever the shape of the errors. A decision maker who picks this way, in their words, "should expect to be disappointed on average, even if the individual value estimates are conditionally unbiased" [2]. They named it the optimizer's curse, after the winner's curse in auctions. They called it "at best underappreciated" in their field [2].

Their proof is for a single choice. On this paper's reasoning, the same logic covers funding several proposals from the top. That is choosing by a line, and the proposals above it share the problem. Ryan Kessler of Amazon showed a matching result for product features launched after A/B tests. There, a feature launches only if its tested result clears a bar [4]. Each test's result is honest on average. But the company "is more likely to launch a feature precisely when frequentist estimates overestimate its impact." So the launched features' results, added up, are too high [4].

Honest means right on average for each proposal before anyone ranks them. It does not mean right on average among the ones that win.

What makes the gap bigger

The size of the gap depends on two things: how many close rivals there are, and how uncertain each estimate is. Smith and Winkler measured the uncertainty as the spread of an estimate's error. That is roughly how far an estimate usually lands from the truth. Take their worst case. Every option is truly worth the same, every estimate has the same spread, and the errors are independent of each other. Then the winner's estimate is too high by about 0.56 of that spread with two options, 0.85 with three, 1.03 with four and 1.54 with ten [2]. Suppose, for illustration, that an estimate usually lands about $1 million away from the truth. Then among ten equally good proposals, the winner's claim would be about $1.5 million too high on average. Among two, it would be about $0.6 million too high.

That is the worst case, not an estimate for any review. The gap shrinks when the options are truly far apart, because then the best one usually wins anyway. With three options one spread apart, the authors' corrected figures put the gap at 0.54 of the spread [2, 3]. The gap also shrinks when all the estimates share the same errors, such as a common assumption. It grows when the options' true values tend to move together. In one four-option setting they tested, making both kinds of link fairly high cut the gap by up to about half. They found that it "remains substantial" [2].

To see what this can do to a reported total, they took 38 decision studies done at Eastman Kodak in the 1990s. They treated Kodak's estimates as the truth and added random error of 5, 10 or 25 percent. The value the studies claimed to add, the best option compared with the average one, came out 14, 39 or 128 percent too high [2]. These are results of a simulation with assumed error levels, not measured outcomes. They show how much error of these sizes can inflate a total once you pick from the top.

The gap is also uneven. An estimate with a wider error has more room to land high. So when a thinly evidenced proposal tops the list, more of its lead is likely to be error. In Smith and Winkler's model, a chosen claim is expected to fall short by the part of its distance above the typical value that its evidence does not support [2]. So for the same claim above the typical value, a thin winner should be expected to fall further short than a well-measured one. Kessler proved a matching result for features launched after A/B tests. The lower a test's statistical power, the more its launched result is inflated on average [4]. Low power can come from a noisy test or from a small true effect.

An AI round can hold many close ideas backed by very different evidence. One has a demo, one a measured pilot, one is already used elsewhere in the business. That is the situation in which the effect is largest. That claim is this paper's reasoning about AI rounds, not a measured finding about them.

Why the usual discount leaves the same winner

The review already discounts. But a cut of the same share to every proposal's ranked figure does not change the order. If every claim is trimmed by 30 percent, the top claim is still the top claim. So a uniform cut can bring the expected gap down to size. It cannot change which proposal wins. Nor can it keep up with the round. To cancel the gap, it would have to grow as more close rivals enter.

Some common discounts do change the order, but for reasons that have nothing to do with how reliable each estimate is. Say proposals are ranked on benefit minus cost. Then trimming benefits alone hits the proposals with larger benefits harder. One discount rate applied to proposals with different timing favors the ones that pay back sooner. A hurdle under a fixed budget can change which set gets funded.

A confidence tier, like RICE's, does reorder by evidence, but crudely. It pulls every estimate toward zero, not toward what such proposals usually deliver. And its size is one of three fixed steps (high, medium or low). It is not a judgment of how much this proposal's evidence narrows the range. A room that uses tiers may already discount a demo hard. RICE calls anything below 50 percent a "total moonshot" [1].

Smith and Winkler showed what does change the winner. Their fix moves each estimate part of the way toward a typical value. The share it keeps is set by how reliable it is. When every estimate is equally reliable and every proposal starts from the same typical value, "the ranking of alternatives is not changed" [2]. When reliability differs, "alternatives whose values are hard to estimate may be passed over by alternatives with lower, but more accurate, value estimates" [2]. The fix also raises low estimates toward the typical value. So it is not simply a penalty for uncertainty. They add that experienced decision makers who insist on high hurdles may be showing "(in part) a learned response to the effects of the curse" [2]. Your instinct to discount is sound. The form it takes decides whether it changes anything.

What Amazon found in its own tests

Does the correction change anything in a company's own data? Kessler applied the same kind of correction to Amazon's A/B tests from 2023 [4]. He notes that "many data-driven companies measure the impact of product groups and allocate resources across them." They do it by adding up the tested effects of the features each group launches [4]. He re-analyzed the tests under a standard launch rule he assumed for simplicity, using a disguised engagement measure.

After the correction, the estimated total impact of the launched features was 0.413 of the plain sum, "nearly 60 percent smaller" [4]. The cut was not even across product groups. One group's share of the total rose by about 40 percent. Another's fell by about half. On

this paper's reading of his table, the top two groups stayed in place. On the same reading, one group moved from sixth, behind a three-way tie, to third [4]. The plain sum, he concludes, "can significantly overstate the impact of product groups and distort the allocation of resources" [4].

The limits matter here. This is one company, one year and a disguised measure. The tests are many small, similar experiments, not a few very different proposals. Features launch by clearing a bar, not by beating rivals. So the study supports the reordering, not the growth with the size of the round. And the corrected figures are estimates from a model, not measured results. Moving from product features to investment proposals is an analogy. What carries over is the structure: honest estimates of different precision, chosen by a line, overstated unevenly. In Amazon's case the correction did not change which features launched. It changed how the product groups ranked. So it would change which groups a budget line reached, if resources followed those totals.

How to move the figures

The correction needs judgments a chair can give, not a data team. Smith and Winkler suggest asking, "Before you saw the results of this analysis, what would you have estimated the value to be?" [2]. Records of how similar AI proposals turned out are rare. Smith and Winkler found the same for decision analyses in general [2]. So the practical starting point is the chair's own typical figure and range. The chair writes them down before reading the cases, so the cases cannot move them. The typical figure is for proposals of this kind and size. Where proposals differ widely in scope, set a typical figure for each kind.

Then, for each proposal, judge how far its evidence narrows that range. The moved figure is the typical figure plus a share of the distance between the claim and the typical figure. The more the evidence narrows the range, the more of the distance the claim keeps. That share is the claim's weight. In Smith and Winkler's method, the weight is the part of the uncertainty the evidence removes, measured as variance [2]. In plain terms: take the fraction of the range's width that is left after the evidence. Square it. Subtract the result from one. Halving the range leaves half its width, so the claim keeps 75 percent. Narrowing it by a tenth keeps 19 percent. Narrowing it by 60 percent keeps 84 percent, and by 80 percent, 96 percent. The authors mention keeping 80 to 95 percent only as "a representative range" for formal decision analyses [2]. There is no default share. It is the chair's judgment of each kind of evidence.

How much does a demo narrow the range? This paper's reasoning is that it narrows it only a little. A demo shows the tool can do the task on the cases shown. The value depends on things a demo does not measure: how much work flows through the task,

whether people adopt the tool, how errors are handled and what it costs to run. A measured pilot narrows the range more. Use elsewhere in the business narrows it more again.

Here is a worked illustration. The numbers are chosen to show the method, not taken from any review. The chair's typical figure is $1.0 million a year.

A demo-only proposal claims $4.0 million. The chair judges the demo narrows the range by a tenth. So it keeps 19 percent of its distance from the typical figure. Moved figure: $1.57 million.

A measured pilot claims $2.4 million. It narrows the range by 60 percent, so it keeps 84 percent. Moved figure: $2.18 million.

A tool already in use elsewhere claims $2.0 million. It narrows the range by 80 percent, so it keeps 96 percent. Moved figure: $1.96 million.

On the moved figures, the pilot ranks first, then the tool in use, then the demo. On the raw claims, the demo is first. Cut all three by 30 percent ($2.8, $1.68 and $1.4 million), and the demo is still first. Now use RICE-style tiers of 50, 80 and 100 percent ($2.0, $1.92 and $2.0 million). The demo and the tool in use tie for first, and the pilot comes last. The demo stays tied for first because the 50 percent tier leaves it $2.0 million. The chair's judgment, by contrast, leaves it $1.57 million, about 39 percent of its claim. The pilot drops because the tier pulls toward zero rather than toward the typical figure.

The result is sensitive to the chair's judgment, as it should be. Suppose the chair judged that the demo narrowed the range by more than about 22 percent, keeping more than about 39 percent of its distance. Then the demo would rank first on the moved figures too. The illustration shows the method. It is not a recommended share.

Two objections attack this step. The first: "The typical figure is subjective." It is a judgment, which is why it is written down before the cases are read. Ranking on raw claims is also a judgment: that every estimate is equally reliable. Kessler's simulations found the correction still made the error in the total smaller when the spread of true effects was misjudged. That held unless the spread was understated by more than about twofold [4]. Those simulations kept the typical value right. So they do not show what a wrong typical figure does. That is why delivery is checked against the moved figures, where a typical figure set wrong would show up.

The second: "This kills bold ideas." The adjustment moves claims toward the typical figure from both sides. It cuts thin high claims, and it raises thin low ones. For a new kind of work, the chair's own range is wide. The same evidence then removes a larger share of that range than it would of a narrow one. So the claim keeps more of its weight. A bold claim with thin evidence will rank lower than its raw claim, as it should. A bold idea with strong evidence still ranks well. And a funded idea's moved figure is what the business expects, not a cap on what it can deliver.

Reading a shortfall

When a funded proposal falls short of its claim, the usual reading is that its sponsor oversold. But the ranking produces a shortfall even when no one oversold. Overselling and selection both leave winners below their claims. So a shortfall against the claim cannot tell them apart. Where estimates are also optimistic, Smith and Winkler note, "the expected disappointment will be even greater" [2].

The moved figure helps here. Smith and Winkler showed that an option chosen on properly adjusted figures carries no expected shortfall against its adjusted figure [2]. So the portfolio office records both the claim and the moved figure at approval. Delivery is judged against the moved figure at the first review at which the benefit can be measured. A shortfall against the raw claim is what the ranking predicts. Now suppose shortfalls against the moved figure recur across many funded proposals. That points to something else. It may be optimism, or a typical figure set wrong. It may be weights set too high because the evidence was judged to narrow the range more than it did. Or it may be change after approval.

On this paper's reasoning, those causes leave different traces. Optimism survives the adjustment mostly in well-measured winners, because they keep most of their claims. A typical figure or weights set wrong show mostly in thinly evidenced winners, whose moved figures rest mostly on those judgments. Two simpler comparisons mislead. One compares crowded rounds with quiet ones. Proposals in crowded rounds may be oversold more, because competition invites it. The other compares contested proposals with those approved without rivals. Those approved without rivals still had to pass a bar, and they may be a different kind of work.

This is a reading across many proposals. No single shortfall is a verdict on a sponsor. Smith and Winkler say documenting the curse from real outcomes is hard. They give two reasons. Few firms keep records of their estimates, and actual results are hard to pin down and separate from other causes [2]. Recording the claim and the moved figure at approval builds that record.

The objections

"We already apply a confidence factor." Keep the instinct. But the factor pulls toward zero, not toward what such proposals deliver. So the figure it leaves is not one that delivery can fairly be judged against, and a shortfall still looks like overselling.

"Our sponsors do oversell." Then both forces act, and the gap is larger. Nothing here assumes honesty. It shows the gap appears even with honesty.

"Our analysts know the winner's curse from A/B testing." Then they will recognize it here. The same logic applies to the proposals they bring to the portfolio review.

What to change in the review

Make three changes to how the review ranks proposals. First, before reading the cases, the chair writes down a typical figure for this kind of proposal and the range they would expect. Second, for each proposal, the chair judges how far its evidence narrows that range. The chair then moves the proposal's claim toward the typical figure by the matching share, keeping more where the evidence narrows the range more. Third, rank and fund on the moved figures. Where the review ranks on value for the effort, divide the moved figure, not the claim, by the effort.

If the typical figure and range are about right, the moved figures need no extra allowance for a crowded round. In that case, they already reflect how far each estimate can be trusted. The portfolio office records the claim and the moved figure at approval. Delivery is judged against the moved figure when the benefit can first be measured. Gaps against the moved figures that recur across many proposals are a reason to revisit the typical figure and the weights.

The argument has limits. It rests on a mathematical result and simulations, and on one company's re-analyzed product tests. Carrying it to AI investment proposals is an analogy. It does not say how large the gap is in any particular review. It gives no default weight. It covers proposals competing for the same capacity and compared on estimated value or a score built from it. It does not cover commitments where another party can decline the deal, or whether a pilot's result survives scaling up.

The question for the review changes from "which proposal claims the most?" to "which proposal is worth the most once each claim is moved toward what such proposals deliver, by more where its evidence is thinner?"

Sources

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